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Practical - 6



The image shows a Visual Studio Code editor with a Python file named `main.py` open. The code defines a FastAPI application with two routers: `catalog_router` and `admin_router`. The application is titled "BookFlow Library API". The terminal at the bottom shows the command `uvicorn main:app --reload` being executed, with output indicating the server is running on `http://127.0.0.1:8000` and showing successful responses for `GET /docs HTTP/1.1` and `GET /openapi.json HTTP/1.1`.

```
1 from fastapi import FastAPI
2
3 from routes.catalog import router as catalog_router
4 from routes.admin import router as admin_router
5
6
7 app = FastAPI(
8     title="BookFlow Library API"
9 )
10
11
12 app.include_router(catalog_router)
13 app.include_router(admin_router)
```

```
PS C:\Users\birad\OneDrive\Documents\DB&BE Complete Project\BookFlowp6> uvicorn main:app --reload
BookFlowp6]
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [10152] using StatReload
INFO:     Started server process [24964]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     127.0.0.1:50060 - "GET /docs HTTP/1.1" 200 OK
INFO:     127.0.0.1:61494 - "GET /openapi.json HTTP/1.1" 200 OK
```

Swagger:

BookFlow Library API 0.1.0 OAS 3.1

/openapi.json

default

GET	/books	Get Books
GET	/books/{book_id}	Get Book
POST	/admin/books	Add Book
PUT	/admin/books/{book_id}	Update Book
DELETE	/admin/books/{book_id}	Delete Book

Schemas

Book > Expand all object

HTTPValidationError > Expand all object

ValidationError > Expand all object

GET/books:

default

GET

/books

Get Books

^

Parameters

Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header

Example Value | Schema

"string"

GET

/books

Get Books

^

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

curl -X 'GET' \n'http://127.0.0.1:8000/books' \n-H 'accept: application/json'

Request URL

http://127.0.0.1:8000/books

Server response

Code	Details
200	Response body <pre>{ "id": 301, "title": "Python Programming", "author": "John Smith", "price": 450, "isbn": "9781234567890", "status": "Available" }, { "id": 302, "title": "Database Management", "author": "Robert Brown", "price": 550, "isbn": "9789876543210", "status": "Available" }]</pre> Download

Books_id:

GET

/books/{book_id} Get Book

⌵

Parameters

Cancel

Name	Description
book_id required	
integer	
(path)	

Execute

Responses

Code	Description	Links
200	Successful Response	No links
	Media type	
	application/json	
	Controls Accept header.	
	Example Value Schema	
	"string"	
422	Validation Error	No links

Execute

Clear

Responses

Curl

curl -X 'GET' \

'http://127.0.0.1:8080/books/101' \

-H 'accept: application/json'

Request URL

http://127.0.0.1:8080/books/101

Server response

Code	Details
200	Response body
	{
	"id": 101,
	"title": "Python Programming",
	"author": "John Smith",
	"price": 450,
	"isbn": "9781234567890",
	"status": "Available"
	}
	Response headers
	content-length: 117
	content-type: application/json
	date: Fri, 18 Sep 2026 03:43:26 GMT
	server: unicorn

Responses

GET

/books/{book_id} Get Book

⌵

Parameters

Cancel

Name	Description
book_id required	
integer	
(path)	

Execute

Clear

Responses

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/books/102' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/books/102
```

Server response

Code	Details
200	Response body <pre>{ "id": 102, "title": "Database Management", "author": "Robert Brown", "price": 550, "isbn": "9789876543210", "status": "Available" }</pre> Download Response headers <pre>content-length: 128 content-type: application/json date: Fri, 18 Sep 2026 03:44:02 GMT server: uvicorn</pre>

Responses

Post:

POST /admin/books Add Book

Parameters

CancelReset

No parameters

Request body required

application/json

Edit ValueSchema

```
{
  "title": "Operating Systems",
  "author": "Galvin",
  "price": 600,
  "isbn": "9781111111111",
  "status": "Available"
}
```

Execute

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/admin/books' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Operating Systems",
    "author": "Galvin",
    "price": 600,
    "isbn": "9781111111111",
    "status": "Available"
  }'
```

Request URL

http://127.0.0.1:8000/admin/books

Server response

Code	Details
200	Response body <pre>{ "message": "Book added successfully", "book": { "id": 103, "title": "Operating Systems", "author": "Galvin", "price": 600, "isbn": "9781111111111", "status": "Available" } }</pre> Response headers <pre>content-length: 189 content-type: application/json date: Fri, 18 Sep 2026 03:45:32 GMT server: uvicorn</pre>

PUT:

PUT /admin/books/{book_id} Update Book

Parameters

Name	Description
book_id ^{required}	
integer	
(path)	

Request body ^{required}

application/json

Edit Value | Schema

```
{
  "title": "Advanced Operating Systems",
  "author": "Galvin",
  "price": 650,
  "isbn": "9781111111111",
  "status": "Available"
}
```

Execute

Responses

Execute

Clear

Responses

Curl

```
curl -X 'PUT' \
  http://127.0.0.1:8000/admin/books/103' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Advanced Operating Systems",
    "author": "Galvin",
    "price": 650,
    "isbn": "9781111111111",
    "status": "Available"
  }'
```

Request URL

http://127.0.0.1:8000/admin/books/103

Server response

Code

Details

200

Response body

```
{
  "message": "Book updated successfully",
  "book": {
    "id": 103,
    "title": "Advanced Operating Systems",
    "author": "Galvin",
    "price": 650,
    "isbn": "9781111111111",
    "status": "Available"
  }
}
```

Download

Response headers

```
content-length: 170
content-type: application/json
date: Fri, 18 Sep 2020 03:47:44 GMT
server: uvicorn
```

DELETE:

DELETE

/admin/books/{book_id}

Delete Book

Parameters

Cancel

Name	Description
book_id * required	
integer	
(path)	

Execute

Responses

Code	Description	Links
200	Successful Response	No links
	Media type application/json Controls Accept header. Example Value Schema "string"	
422	Validation Error	No links
	Media type	

Execute

Clear

Responses

Curl

```
curl -X 'DELETE' \
'http://127.0.0.1:8000/admin/books/101' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/admin/books/101
```

Server response

Code

Details

200

Response body

```
{
  "message": "Book deleted successfully",
  "book": {
    "id": 101,
    "title": "Python Programming",
    "author": "John Smith",
    "price": 450,
    "isbn": "9781234567890",
    "status": "Available"
  }
}
```

Download

Response headers

```
content-length: 164
content-type: application/json
date: Fri, 19 Sep 2020 03:48:29 GMT
server: uvicorn
```

Responses